

ATPL EASA STUDY GUIDE | 021 - AIRFRAME & SYSTEMS

Chapter 21: Combustion, Turbines & Exhaust / Reversers

Syllabus Source: CAE Oxford (pp. 195-256 (Pwr)) | Pass Target: >= 90% AviationExam

1. COMBUSTION CHAMBER ARCHITECTURE & AIR DISTRIBUTION

Combustion chambers burn fuel continuously to add thermal energy at constant pressure. Air entering from the compressor (at ~150 m/s) is slowed down by a diffuser to ~25 m/s to prevent blowing the flame out (flame speed is only ~2 to 5 m/s!):

Airflow Division	Percentage of Total Bleed Air	Operational Combustion Function
Primary Air	20% to 25%	Enters flame tube snout through swirl vanes to create a toroidal recirculation vortex. Mixes stoichiometrically (~15:1 air-fuel ratio) to support sustained ignition and stable core burning.
Secondary Air	15% to 20%	Enters through intermediate liner holes to complete combustion of unburned hydrocarbons.
Tertiary / Dilution Air	55% to 60% (The Majority!)	Enters downstream liner holes. Cools burning gases from ~2,000°C down to acceptable turbine entry temperatures (~1,200°C to 1,600°C) and forms an insulating film along chamber walls.

2. AXIAL FLOW TURBINES & BLADE COOLING

Turbines extract kinetic and heat energy from hot expanding combustion gases to drive the compressor and engine accessories:

- Nozzle Guide Vanes (NGVs): Stationary vanes positioned immediately upstream of the turbine rotor. Passages form CONVERGENT DUCTS that accelerate the gas to high velocity and deflect it at the optimum angle onto the turbine blades. Experiences highest thermal and aerodynamic stresses in the entire engine!
- Impulse vs Reaction Blading: Impulse: Gas pressure drops exclusively across NGVs, expanding as high-speed jets against curved blades. Reaction: Pressure drops across BOTH the NGVs and the rotating turbine blades (convergent blade passages). Modern engines use impulse blade roots transitioning to reaction blade tips.
- Turbine Blade Cooling: Blades are cast from single-crystal nickel superalloys. Cooled by compressor bleed air: 1. Internal convection passages; 2. Transpiration/film cooling where bleed air bleeds out through laser-drilled leading-edge holes, forming a protective microscopic thermal air cushion over the blade surface.
- Active Clearance Control (ACC): Engine casing cooling system that blows fan bypass air onto the turbine casing, matching thermal casing shrinkage to turbine rotor growth, maintaining microscopic blade-tip clearances (minimizing gas leakage).

3. EXHAUST SYSTEMS & THRUST REVERSERS

Exhaust nozzles accelerate combustion gases overboard to create forward thrust (Gross Thrust $F = \dot{m} \times V_{jet}$):

- Convergent Nozzle: Used on all commercial subsonic airliners. Accelerates subsonic exhaust gas up to sonic velocity (Mach 1.0) at the nozzle throat (choked nozzle condition).

- Thrust Reversers: Redirect fan bypass air (or core exhaust) forward to provide aerodynamic deceleration on landing: 1. Clamshell / Bucket doors (deflects core exhaust); 2. Cascade Vane / Translating Sleeves (aerodynamic blocker doors block fan duct, directing bypass air forward through cascade vanes). In-flight deployment is physically blocked by multiple mechanical and electrical interlocks.

[!] EXAM TRAP (AVIATIONEXAM) - Thrust Reverser Interlocks & In-Flight Safety

- Deployment on Ground: Requires weight-on-wheels (squat switch confirmation), throttle levers at idle, and reverse levers pulled. - In-Flight Protection: Features dual mechanical latches, hydraulic isolation valves, and auto-restow circuitry. If a reverser unlatches inadvertently in flight (e.g. Lauda Air 004 scenario), FADEC automatically reduces that engine's fuel flow to flight idle!

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Turbine Creep Life & Reverse Interlocks

1. Creep Stages: Primary (decreasing rate), Secondary (slow steady rate during normal service), Tertiary (exponential acceleration to rupture). 2. Creep Prevention: Single-crystal nickel superalloy blades without transverse grain boundaries. 3. Thrust Reverser Safety Interlock: Mechanical balk stops thrust lever movement until blocker doors are fully deployed in reverse position.

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: What percentage of total compressor air entering a gas turbine combustion chamber is used as 'primary air'?

- [A]: 10% to 15%
- [B]: 20% to 25%
- [C]: 50% to 60%
- [D]: 85% to 90%

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Only about 20-25% of the total air is primary air used directly for burning fuel at stoichiometric ratio. The remaining 75-80% is used for combustion completion and cooling dilution.

Q2: What is the primary operational function of turbine Nozzle Guide Vanes (NGVs)?

- [A]: To cool the exhaust gases
- [B]: To accelerate the hot gases through convergent passages and direct them at the optimum angle onto the rotating turbine blades
- [C]: To drive the engine oil pump
- [D]: To reverse engine thrust

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. NGVs are stationary nozzles ahead of the turbine rotor. Their convergent shape converts gas pressure into high-velocity kinetic energy, directing the jet at the ideal impingement angle onto the blades.

Q3: How are modern high-pressure turbine blades protected against melting at temperatures exceeding the all-

- [A]: By thermal paint only
- [B]: By internal convection passages and external film cooling using relatively cool compressor bleed air
- [C]: By injecting water into the fuel
- [D]: By flying only in sub-zero ambient air

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Bleed air from the HP compressor is routed through internal labyrinth passages in the blade and exits through micro-holes to create a boundary film of cooler air shielding the metal from 1,500°C gases.